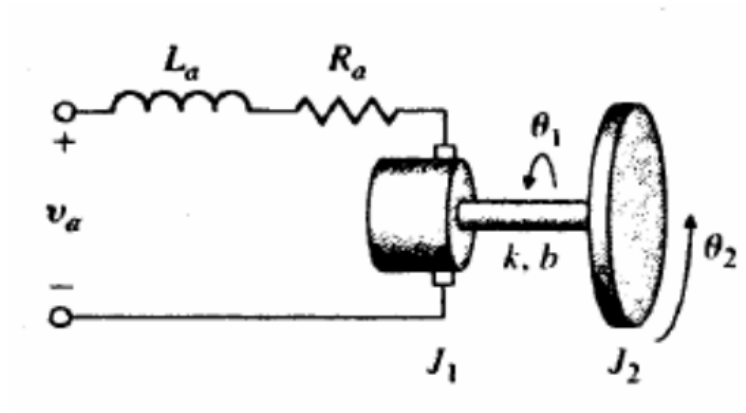


Homework 4

EE 999 — Introduction to Control Systems

Patrick Gordon • October 1, 2026

Due Wednesday, October 7, 2026

Problem 1.A — Equations of Motion
Figure 1: DC motor driving a load inertia J_2

Electrical side:

$$v_a(t) = v_L(t) + v_R(t) + v_{J_1}(t)$$

$$v_L(t) = L_a \dot{i}_a(t)$$

$$v_R(t) = R_a i_a(t)$$

$$v_m(t) = K_e \dot{\theta}_1(t)$$

Motor:

$$M_c \ddot{\theta}_1(t) = T_{net.J_1}(t)$$

$$T_m(t) = k_t i_a(t)$$